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Federated Spatio-Temporal Policy Learning for Efficient Resource Management in UAV-Assisted IoT Edge Networks

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***Abstract*—Federated learning (FL) enables privacy-preserving distributed intelligence in UAV-assisted Internet of Things (IoT) edge networks. However, efficient policy design remains challenging due to the hybrid decision space, which couples combinatorial device scheduling with continuous resource allocation, as well as the long-term impact of non-independent and identically distributed (Non-IID) data on model convergence. In this paper, we propose a spatio-temporal resource optimization framework for UAV-assisted edge computing services. We formulate a long-term utility maximization problem that jointly considers learning performance, energy consumption, and latency. To solve this mixed-integer nonlinear problem, a federated spatio-temporal policy optimization (FSTPO) approach is developed based on a constrained Markov game. The proposed method incorporates a dual-channel spatio-temporal representation to capture wireless dynamics and data distribution characteristics. In addition, a hybrid action learning mechanism is introduced to efficiently handle the coupling between discrete scheduling and continuous resource allocation. Simulation results demonstrate that the proposed approach outperforms benchmark schemes in terms of system utility, learning accuracy, and resource efficiency, while achieving stable convergence under Non-IID environments.**

***Index Terms*—Federated learning, UAV-assisted IoT networks, resource management, spatio-temporal policy learning, edge computing.**

I. Introduction

A. Background

THE large-scale deployment of Internet of Things (IoT) devices has led to the continuous generation of massive and heterogeneous data at the network edge [1], [2]. Under increasingly stringent privacy constraints, federated learning (FL) has emerged as a key paradigm for distributed model training without sharing raw data [3]. However, in infrastructure-limited scenarios, edge devices are often constrained by limited battery capacity and computational power, making it difficult to support independent local model training [4]. In addition, traditional fixed base stations suffer from limited coverage and flexibility. To address these limitations, integrating unmanned aerial vehicles (UAVs) with mobile edge computing (MEC) offers a promising solution, enabling reliable line-of-sight communication and flexible computing support for ground devices, thereby alleviating the deployment bottlenecks of FL in infrastructure-sparse environments [5]–[7].

Despite these advantages, UAV-assisted FL systems introduce significant challenges in both system modelling and resource management. The high mobility of UAVs leads to dynamically evolving air–ground topologies, resulting in time-varying and intermittent service availability [8]. Meanwhile, the limited onboard computation and communication resources intensify competition among devices [9]. Critically, due to differences in sensing environments and task characteristics, local datasets across IoT devices are inherently non-independent and identically distributed (Non-IID), which severely degrades the stability of model aggregation. Given these competing demands, efficient task offloading and resource scheduling become essential for bridging the learning requirements and the available aerial resources [10], [11]. In particular, joint optimization of device association, task offloading, and resource allocation has become a key research focus [12].

Existing work has made notable progress in energy efficiency and latency optimization [7], [13]–[15]. Early studies primarily rely on optimization theory and game-theoretic approaches to address multi-user resource competition. For instance, some works jointly optimize UAV trajectories and transmission power via alternating optimization or convex relaxation to minimize system delay. Heuristic approaches have also been explored for complex scenarios. Tian et al. investigated task offloading and scheduling in UAV-enabled MEC networks to enhance service satisfaction via user grouping and genetic algorithms [16]. Wang et al. proposed a bi-criteria ant colony optimization framework to minimize system cost and completion time [17]. Zhao et al. employed Gibbs sampling and matching theory to maximize quality of experience under limited resources [18]. However, these methods generally rely on static or quasi-static assumptions and cannot capture the dynamic coupling among communication, computation, and learning processes in highly time-varying environments.

With the increasing complexity and non-stationarity of wireless networks, research has increasingly shifted toward deep

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reinforcement learning (DRL) and FL-based joint optimization [19]. DRL provides a promising approach for handling high-dimensional, dynamic decision-making problems, while FL enables distributed model training without sharing raw data. Lee et al. provided a comprehensive survey on FL-enabled network management, highlighting the critical role of artificial intelligence in addressing complex resource allocation challenges [20]. Building upon this paradigm, several studies have explored DRL-driven resource optimization in heterogeneous networks. Khan et al. proposed a federated DRL framework for joint discrete offloading and continuous resource allocation, effectively reducing system cost [21]. Alqahtani et al. introduced the FLARE framework to improve resource utilization through adaptive learning rate scheduling and gradient sparsification [22]. Zhou et al. developed a framework for customized virtual reality (VR) streaming services. At the same time, Seo et al. proposed a carbon-aware FL framework to reduce energy consumption and emissions in cloud environments [23], [24].

In practical edge environments, the coexistence of heterogeneous computation and communication resources further complicates FL deployment. Devices differ significantly in terms of processing capability, bandwidth availability, and energy constraints, which leads to highly uneven participation in distributed training [25]. To address these challenges, researchers have explored various strategies, including split learning and personalized FL. Peng et al. proposed the FedFold framework to improve training efficiency under Non-IID data through progressive model partitioning [26]. Fan et al. developed a multi-agent DRL-based resource management scheme for collaborative cloud-edge-device systems [27]. Qiang et al. applied adaptive parallel split learning to vehicular edge computing scenarios [28], while Kwantwi et al. proposed a personalized FL framework to jointly optimize latency, energy consumption, and personalization [29]. For more dynamic environments, Tang et al. proposed an online resource allocation algorithm that integrates Lyapunov optimization with DRL for UAV-assisted systems [30]. Li et al. developed a distributed offloading scheme for hotspot scenarios, and Karmakar et al. introduced FairLearn to enhance fairness and security in federated systems [31], [32]. Nevertheless, these studies generally treat resource allocation and learning dynamics in a decoupled manner and fail to capture the intrinsic coupling among dynamic scheduling, resource constraints, and Non-IID data distributions. As a result, the long-term impact of scheduling decisions on model convergence remains poorly understood, particularly in UAV-assisted FL systems operating in highly dynamic air–ground environments.

In summary, existing studies largely overlook the long-term coupling between dynamic scheduling decisions and model convergence under Non-IID data distributions [3], [16], [19]. In highly dynamic UAV-assisted environments, scheduling strategies that rely solely on instantaneous physical states may introduce systematic bias by repeatedly excluding statistically important devices, thereby deviating the global model from the optimal convergence trajectory. Therefore, how to jointly design adaptive scheduling and resource allocation mechanisms that balance communication efficiency and learning robustness remains an open and critical problem.

B. Motivation and Contributions

Motivated by the aforementioned discussion, we consider a UAV-assisted FL system in air–ground integrated networks, where mobile edge computing (MEC) servers are mounted on UAVs to support distributed model training and task execution. This design is particularly motivated by scenarios in which fixed ground infrastructure is absent or operationally impractical, such as post-disaster emergency response, remote-area IoT monitoring, and temporary field deployments, where UAVs represent the only viable means of providing flexible, on-demand edge computing and aggregation services. While this architecture provides enhanced flexibility and coverage, the coexistence of dynamic network topology, limited communication/computation resources, and severe Non-IID data distribution poses significant challenges to efficient system design. In particular, the strong coupling among device scheduling, bandwidth allocation, and computation control makes it difficult to achieve stable model convergence and efficient resource utilization simultaneously.

In this paper, we investigate a UAV-assisted federated edge learning framework that jointly optimizes communication, computation, and learning performance under heterogeneous environments. Specifically, we develop a unified spatio-temporal resource optimization framework to maximize system utility while satisfying energy-efficiency, latency, and model-accuracy requirements. The main contributions of this paper are summarized as follows:

- We develop a system model for UAV-assisted federated edge learning that jointly captures device access control, bandwidth allocation, and computation resource adaptation under a dynamic air–ground topology. A system utility function is formulated to characterize the trade-off among FL convergence performance, energy consumption, and latency, while explicitly incorporating data heterogeneity via gradient divergence.
- We propose a federated spatio-temporal policy optimization (FSTPO) framework to address the resulting mixed-integer nonlinear optimization problem. The problem is modelled as a decentralized, constrained Markov game, enabling distributed, scalable decision-making among multiple UAV agents.
- We design a dual-channel spatio-temporal graph-based state representation that integrates physical communication features and semantic information reflecting data heterogeneity. This representation enhances the policy's ability to capture complex correlations and improves robustness against Non-IID data distributions.
- Extensive experiments on Fashion-MNIST and CIFAR-10 under Non-IID settings demonstrate that FSTPO achieves superior utility, accuracy, and resource efficiency over all baselines while satisfying latency and reliability constraints across varying network scales.

The rest of this paper is organized as follows. Section II introduces the UAV-assisted federated learning system model and problem formulation. Section III presents the proposed FSTPO algorithm framework in detail. Section IV describes the experimental settings and parameter configurations. Sec-

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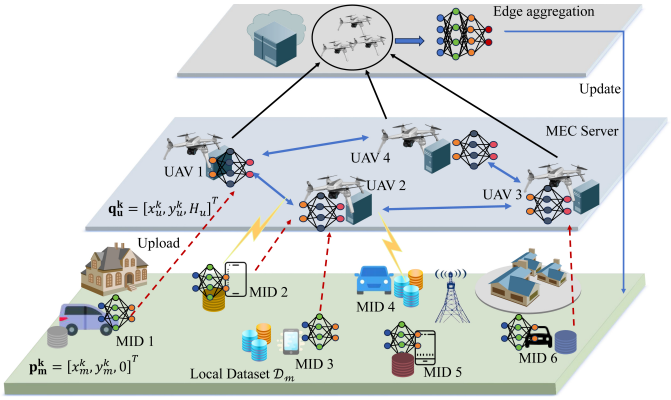


Fig. 1: The proposed UAV-assisted edge computing networks.

tion V provides simulation results. Finally, Section VI concludes the paper and outlines future research directions.

II. System Model and Problem Formulation

A. Network Model

As illustrated in Fig. 1, we consider a UAV-assisted mobile edge computing (MEC) system, where a set of low-altitude unmanned aerial vehicles (UAVs) is denoted by $\mathcal{U} = \{1, 2, \ldots, U\}$, and a set of ground mobile IoT devices (MIDs) is denoted by $\mathcal{M} = \{1, 2, \ldots, M\}$. Each UAV is equipped with an onboard MEC server that serves as an aerial node for both model aggregation and computation offloading. The system operation is discretized into a sequence of global communication rounds indexed by $k \in \mathcal{K} = \{1, 2, \ldots, K\}$. In the k-th round, the three-dimensional position of UAV $u \in \mathcal{U}$ and the horizontal position of device $m \in \mathcal{M}$ are denoted by $\mathbf{q}_u^k = [x_u^k, y_u^k, H_u]^T$ and $\mathbf{p}_m^k = [x_m^k, y_m^k, 0]^T$, respectively. Due to node mobility, the air–ground network topology evolves dynamically across communication rounds, leading to time-varying channel conditions and communication performance. In each communication round, ground devices generate both local training updates and computation tasks. To characterize the device association and task offloading decisions, we introduce a binary scheduling variable $s_{m,u}^k \in \{0, 1\}$. Specifically, $s_{m,u}^k = 1$ indicates that device m is associated with UAV u in round k and offloads its task to UAV u; otherwise, $s_{m,u}^k = 0$ indicates that no connection is established between them.

Given the scheduling decisions, the system operates as follows. At the beginning of each round, devices perform local model training based on their private datasets and generate updated model parameters. Meanwhile, computation tasks—representing latency-sensitive application-layer workloads independent of FL training, such as inference queries and sensor data processing—are also generated at each device. Subsequently, devices establish wireless links with their associated UAVs to upload both local model parameters and task data. Upon receiving the uploaded information, each UAV executes edge computation for the offloaded tasks and aggregates the model updates from associated devices to produce an updated global model.

B. FL Model

Consider a set of ground devices $m \in \mathcal{M}$, where each device locally stores an independent dataset denoted by $\mathcal{D}_m$. Let D_m represent the number of data samples in $\mathcal{D}_m$. For the i-th sample in $\mathcal{D}_m$, its feature vector and corresponding label are denoted by $\mathbf{x}_{m,i}$ and $y_{m,i}$, respectively, where $i \in \{1, 2, \ldots, D_m\}$. The empirical loss function of device m over its local dataset is defined as

$$L_m(\mathbf{w}) = \frac{1}{D_m} \sum_{i=1}^{D_m} \ell(\mathbf{w}; \mathbf{x}_{m,i}, y_{m,i}), \tag{1}$$

where $\ell(\cdot)$ denotes the loss function of a single sample, and $\mathbf{w} \in \mathbb{R}^d$ represents the global model parameter vector with dimension d. The global loss function is defined as

$$F(\mathbf{w}) = \sum_{m \in \mathcal{M}} \frac{D_m}{D_{\text{total}}} L_m(\mathbf{w}), \tag{2}$$

where $D_{\text{total}} = \sum_{m \in \mathcal{M}} D_m$ denotes the total number of data samples across all devices.

During the k-th communication round, only the devices with scheduling indicator $s_{m,u}^k = 1$ are selected to participate in training. Each selected device first receives the global model $\mathbf{w}^{k-1}$ via the downlink and performs local updates initialized from it. The local stochastic gradient descent (SGD) update is expressed as

$$\mathbf{w}_m^{(k,i)} = \mathbf{w}_m^{(k,i-1)} - \eta \hat{\nabla} L_m(\mathbf{w}_m^{(k,i-1)}), \quad i \in \{1, \ldots, I\}, \tag{3}$$

where η is the learning rate, $\hat{\nabla} L_m(\cdot)$ is the stochastic gradient computed from sampled local data, and I denotes the number of local iterations.

After completing I iterations, device m obtains its updated local model $\mathbf{w}_m^k = \mathbf{w}_m^{(k,I)}$, which is then uploaded to its associated UAV.

Upon receiving local models, each UAV $u \in \mathcal{U}$ performs cluster-wise aggregation over the associated devices to generate an edge model $\mathbf{w}_u^k$. Subsequently, global aggregation is conducted across all UAVs by a ground-based central aggregation server via a dedicated backhaul link, yielding the updated global model

$$\mathbf{w}^k = \frac{\sum_{u \in \mathcal{U}} \left(\sum_{m \in \mathcal{M}} s_{m,u}^k D_m \right) \mathbf{w}_u^k}{\sum_{u \in \mathcal{U}} \sum_{m \in \mathcal{M}} s_{m,u}^k D_m}. \tag{4}$$

The updated global model $\mathbf{w}^k$ is then broadcast back to all UAVs at the beginning of the next round for downlink distribution to selected devices.

Due to the heterogeneity of local environments and task characteristics, the data distributions across devices are generally non-IID, which leads to deviations between local gradients and the global optimization direction. To characterize this effect, we introduce the gradient divergence metric. Specifically, the squared upper bound of the deviation between the local gradient of device m and the global gradient is defined as

$$(\delta_m^{k-1})^2 = \left\| \nabla L_m(\mathbf{w}^{k-1}) - \nabla F(\mathbf{w}^{k-1}) \right\|^2, \tag{5}$$

where $\nabla L_m(\cdot)$ and $\nabla F(\cdot)$ denote the true local and global gradients, respectively. A larger value of $(\delta_m^{k-1})^2$ indicates

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a greater discrepancy between the local data distribution of device m and the global data distribution.

Due to the limited computation capability of UAVs and constrained wireless bandwidth, it is impractical to schedule all devices in each round. Therefore, the scheduling policy must balance channel conditions and data quality. To this end, we define the federated learning accuracy utility in the k-th round as

$$\Upsilon_{FL}^{k} = \sum_{u\in\mathcal{U}} \sum_{m\in\mathcal{M}} s_{m,u}^{k} \frac{D_m}{D_{\text{total}}} \exp\left(-\lambda(\delta_m^{k-1})^2\right), \tag{6}$$

where $\lambda > 0$ is a penalty coefficient controlling the sensitivity of the utility to gradient divergence. This formulation assigns higher utility to scheduling decisions that prioritize devices with low gradient divergence, providing a differentiable surrogate that guides the policy toward selections beneficial for global model convergence under Non-IID data distributions. To ensure the learning performance, we further introduce the global model accuracy metric Acc^k, which evaluates the prediction accuracy of $\mathbf{w}^k$ on an independent validation dataset. After K communication rounds, the final model is required to satisfy $Acc^K \geq Acc_{\text{target}}$, where Acc_{target} denotes the predefined accuracy threshold.

C. Communication Model

In the k-th communication round, once the scheduling decision is determined, each selected ground device uploads two types of data to its associated UAV: the locally updated model parameters $\mathbf{w}_m^k$ for federated learning, and the computation task data for offloading. Let Z denote the data size of the compressed local model, reflecting the output of a standard compression pipeline including fixed-point quantization and gradient sparsification applied uniformly across all devices. Meanwhile, let v_m^k represent the size of the computation task generated by device m in the k-th round. Accordingly, the total amount of data to be transmitted by device m is given by $\Omega_m^k = Z + v_m^k$.

A frequency division multiple access (FDMA) scheme is adopted, in which each UAV u is assigned an independent spectrum pool of total bandwidth $B_{\max}$, with non-overlapping allocations across UAVs to eliminate inter-UAV co-channel interference. Based on the system geometry, the Euclidean distance between device m and UAV u in the k-th round is expressed as [33]

$$d_{m,u}^{k} = \sqrt{(x_u^k - x_m^k)^2 + (y_u^k - y_m^k)^2 + H_u^2}, \tag{7}$$

Given that low-altitude UAV air-to-ground links typically exhibit a high probability of line-of-sight (LoS) transmission, the free-space path-loss model is adopted. Accordingly, the channel gain between device m and UAV u is given by

$$h_{m,u}^{k} = \frac{\beta_0}{(d_{m,u}^{k})^2}, \tag{8}$$

where β_0 denotes the channel power gain at a reference distance. According to Shannon's theorem, the achievable uplink transmission rate from device m to UAV u is given by

$$R_{m,u}^{k} = b_{m,u}^{k} \log_2\left(1 + \frac{P_m h_{m,u}^{k}}{N_0 b_{m,u}^{k}}\right), \tag{9}$$

where $b_{m,u}^k \geq 0$ denotes the bandwidth allocated by UAV u to device m, P_m is the transmit power of device m, and N_0 represents the power spectral density of additive white Gaussian noise.

D. Energy Model

In this section, we characterize the energy consumption of the air–ground collaborative system in the k-th communication round. The total energy consumption comprises the local computation and transmission energy of ground devices, as well as the edge-computation energy of UAVs.

For each ground device $m \in \mathcal{M}$, energy consumption arises from local federated learning training and data uploading. Only the scheduled devices participate in the above operations, which are controlled by the binary variable $s_{m,u}^k$. Specifically, if device m is scheduled, it performs I local iterations. Let c_m denote the required CPU cycles per bit and f_m^k denote the CPU frequency. The local computation delay and energy consumption are given by $t_m^{\text{comp},k} = ID_m c_m / f_m^k$ and $E_m^{\text{comp},k} = \kappa_m (f_m^k)^2 (ID_m c_m)$, respectively. The communication energy consumption for uploading data to UAV u is given by $E_{m,u}^{\text{comm},k} = s_{m,u}^k P_m \, \Omega_m^k / R_{m,u}^k$ [19].

For each UAV $u \in \mathcal{U}$, the received task v_m^k is processed by its onboard MEC server. Let c_u denote the required CPU cycles per bit of task data and f_u denote the aggregate multi-core computational throughput of the UAV processor. The edge computation delay and energy consumption are given by $t_{m,u}^{\text{edge},k} = v_m^k c_u / f_u$ and $E_{u,m}^{\text{edge},k} = \kappa_u (f_u)^2 (v_m^k c_u)$, which are incurred only when $s_{m,u}^k = 1$. UAV propulsion energy during hovering is not included, as it is independent of the scheduling and resource allocation decision variables and does not affect the relative comparison among methods. To ensure synchronization, the system execution time is defined as

$$T^k = \max_{m\in\mathcal{M},\, u\in\mathcal{U}} \left\{ s_{m,u}^k \left(t_m^{\text{comp},k} + \Omega_m^k / R_{m,u}^k + t_{m,u}^{\text{edge},k} \right) \right\}. \tag{10}$$

For notational simplicity, we define the total per-device service time as $t_{m,u}^k = t_m^{\text{comp},k} + \Omega_m^k / R_{m,u}^k + t_{m,u}^{\text{edge},k}$, so that $T^k = \max_{m\in\mathcal{M},u\in\mathcal{U}} \{ s_{m,u}^k t_{m,u}^k \}$. We further define the per-device energy consumption as $E_{m,u}^k = E_m^{\text{comp},k} + E_{m,u}^{\text{comm},k} + E_{u,m}^{\text{edge},k}$. Accordingly, the total system energy consumption in the k-th round is given by

$$E_{\text{total}}^k = \sum_{u\in\mathcal{U}} \sum_{m\in\mathcal{M}} s_{m,u}^k E_{m,u}^k. \tag{11}$$

E. Reliability Model

In this section, we characterize communication reliability by accounting for small-scale fading. An outage-based model is adopted to quantify the probability of successful transmission. Specifically, the instantaneous channel power gain $\tilde{h}_{m,u}^k$ is assumed to follow an exponential distribution with mean $h_{m,u}^k$.

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Based on this, the instantaneous signal-to-noise ratio (SNR) at UAV u can be expressed as [15]

$$\gamma_{m,u}^{k} = \frac{P_m \tilde{h}_{m,u}^{k}}{N_0 b_{m,u}^{k}}, \tag{12}$$

where $\gamma_{m,u}^{k}$ represents the instantaneous SNR.

To ensure successful decoding, the instantaneous SNR must exceed a predefined threshold γ_{th}. Otherwise, a transmission outage occurs. Accordingly, the communication success probability of device m associated with UAV u in the k-th round is denoted by $\Gamma_{m,u}^{k}$, which can be written as

$$\Gamma_{m,u}^{k} = \Pr\left(\gamma_{m,u}^{k} \geq \gamma_{\mathrm{th}}\right) = \exp\left(-\frac{\gamma_{\mathrm{th}} N_0 b_{m,u}^{k}}{P_m h_{m,u}^{k}}\right), \tag{13}$$

where γ_{th} denotes the SNR threshold for successful decoding.

F. Problem Formulation

In the considered system, ground devices perform local training and simultaneously upload both model parameters and computation tasks to their associated UAVs. The UAVs further execute edge computation and aggregation. Accordingly, the system incurs energy consumption and latency in each communication round while achieving a certain level of learning performance. To jointly evaluate the tradeoff between model accuracy and resource consumption, we define the system utility in the k-th round as

$$U^{k} = \omega_1 \Upsilon_{FL}^{k} - \omega_2 E_{\mathrm{total}}^{k} - \omega_3 T^{k}, \tag{14}$$

where ω_1, ω_2, and ω_3 are positive weighting coefficients that balance the importance of learning performance, energy consumption, and latency, respectively. The weighted-sum formulation parameterizes the Pareto-optimal trade-offs among the three objectives, with coefficients calibrated to bring the terms into a comparable numerical scale.

Our objective is to maximize the cumulative utility over the entire training process by jointly optimizing device scheduling, bandwidth allocation, and CPU frequency control, subject to system resource and reliability constraints. The problem is formulated as

$$\begin{aligned}
&\max_{s_{m,u}^{k},\, b_{m,u}^{k},\, f_{m}^{k}} \quad \sum_{k=1}^{K} U^{k} \\
\text{s.t.} \quad &\text{(C1)}\ s_{m,u}^{k} \in \{0,1\}, \quad \forall m,u,k, \\
&\text{(C2)}\ \sum_{u\in\mathcal{U}} s_{m,u}^{k} \leq 1, \quad \forall m,k, \\
&\text{(C3)}\ \sum_{m\in\mathcal{M}} s_{m,u}^{k} b_{m,u}^{k} \leq B_{\max}, \quad \forall u,k, \\
&\text{(C4)}\ b_{m,u}^{k} \geq 0, \quad \forall m,u,k, \\
&\text{(C5)}\ f_{m}^{\min} \leq f_{m}^{k} \leq f_{m}^{\max}, \quad \forall m,k, \\
&\text{(C6)}\ s_{m,u}^{k}\, t_{m,u}^{k} \leq T_{\max}, \quad \forall m,u,k, \\
&\text{(C7)}\ \Gamma_{m,u}^{k} \geq \Gamma_{\min}, \quad \forall m,u,k, \\
&\text{(C8)}\ Acc^{K} \geq Acc_{\mathrm{target}}.
\end{aligned} \tag{15}$$

The above optimization problem aims to determine the optimal device scheduling $s_{m,u}^{k}$, bandwidth allocation $b_{m,u}^{k}$, and CPU frequency f_{m}^{k} to balance learning performance and resource consumption. Specifically, constraints (C1) and (C2) ensure that each device can be associated with at most one UAV in each communication round. Constraints (C3) and (C4) regulate the bandwidth allocation such that the total allocated bandwidth at each UAV does not exceed its capacity $B_{\max}$ while remaining non-negative. Constraint (C5) ensures that the CPU frequency operates within feasible hardware limits. Constraint (C6) guarantees that the latency of scheduled devices does not exceed the maximum allowable delay $T_{\max}$. Constraint (C7) enforces reliable communication by requiring the success probability to exceed a predefined threshold $\Gamma_{\min}$. Finally, constraint (C8) ensures that the trained global model achieves the target accuracy Acc_{target} after K communication rounds. The formulated problem is a mixed-integer nonlinear programming (MINLP) problem due to the coupling between binary scheduling variables and continuous resource allocation variables, making it intractable for conventional optimization methods.

III. The FSTPO Framework

To address the resulting MINLP problem, where discrete scheduling variables $s_{m,u}^{k}$ are tightly coupled with continuous resource variables $b_{m,u}^{k}$ and f_{m}^{k}, conventional DRL approaches typically rely on discretization or relaxation techniques, which may lead to significant performance degradation in hybrid action spaces.

To overcome this limitation, we propose a Federated Spatio-Temporal Policy Optimization (FSTPO) framework. As illustrated in Fig. 2, the system is modelled as a decentralized constrained Markov game, where each UAV agent independently learns its policy via an actor–critic architecture. The Actor network generates feasible hybrid actions directly from high-dimensional spatio-temporal states, while the Critic network evaluates long-term cumulative rewards to guide policy updates. A federated learning mechanism is incorporated to enable collaborative policy sharing among UAV agents, thereby improving overall system performance under distributed decision-making constraints. The proposed framework adopts an offline training and online deployment paradigm: the RL policy is trained to convergence in a simulated environment prior to deployment, after which each UAV executes only the forward pass of the trained Actor network during online FL operation, imposing negligible inference overhead.

A. Dual-Channel Spatio-Temporal Network

Due to the strong coupling among dynamic wireless channels, device topology, and data heterogeneity, conventional vector-based state representations are insufficient to capture the structured correlations among multi-dimensional features. To address this, we design a dual-channel spatio-temporal perception network as a shared state encoder for both the Actor and Critic networks.

At the k-th communication round, for each UAV $u \in \mathcal{U}$, a local physical topology graph $\mathcal{G}_{\mathrm{phy}}^{k}$ is constructed based on its

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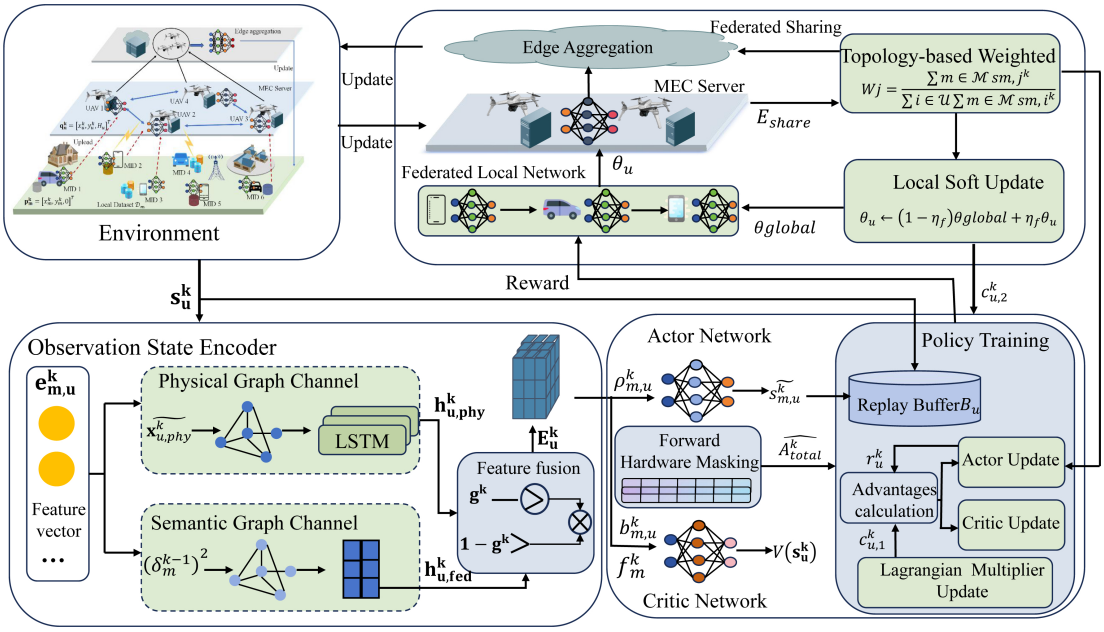


Fig. 2: The structure of the proposed FSTPO Framework.

associated device set $\mathcal{N}(u)$. The edge features are defined by the large-scale channel gain and Euclidean distance as

$$\mathbf{e}_{m,u}^k = [h_{m,u}^k,\, d_{m,u}^k]. \tag{16}$$

A graph attention mechanism is then employed to aggregate neighbourhood information, yielding the spatial feature representation

$$\tilde{\mathbf{x}}_{u,\text{phy}}^k = \text{ReLU}\left(\sum_{m\in\mathcal{M}} \alpha_{m,u}^k \mathbf{W}_p \left[\mathbf{x}_m^k \parallel \mathbf{e}_{m,u}^k\right]\right), \tag{17}$$

where $\mathbf{x}_m^k$ denotes the input feature of device m, $\alpha_{m,u}^k$ is the attention coefficient, and $\mathbf{W}_p$ is a learnable weight matrix.

To capture temporal dynamics induced by time-varying channels and small-scale fading, the spatial features are further processed by a single-layer unidirectional long short-term memory (LSTM) module with hidden dimension $d_h = 128$. The update process is expressed as

$$\mathbf{f}_u^k = \sigma\left(\mathbf{W}_{xf}\tilde{\mathbf{x}}_{u,\text{phy}}^k + \mathbf{W}_{hf}\mathbf{h}_{u,\text{phy}}^{k-1} + \mathbf{b}_f\right), \tag{18}$$

$$\mathbf{C}_{u,\text{phy}}^k = \mathbf{f}_u^k \odot \mathbf{C}_{u,\text{phy}}^{k-1} + \mathbf{i}_u^k \odot \tanh\left(\mathbf{W}_{xc}\tilde{\mathbf{x}}_{u,\text{phy}}^k + \mathbf{W}_{hc}\mathbf{h}_{u,\text{phy}}^{k-1} + \mathbf{b}_c\right), \tag{19}$$

where $\mathbf{h}_{u,\text{phy}}^{k-1}$ and $\mathbf{C}_{u,\text{phy}}^{k-1}$ denote the hidden state and cell state from the previous round, respectively. $\mathbf{f}_u^k$ and $\mathbf{i}_u^k$ represent the forget gate and input gate, $\sigma(\cdot)$ is the sigmoid function, and $\odot$ denotes element-wise multiplication.

In parallel, to capture the impact of data heterogeneity in federated learning, a semantic graph $\mathcal{G}_{\text{fed}}^k$ is constructed based on the gradient divergence $(\delta_m^{k-1})^2$. A graph convolution operation is applied to extract semantic features, denoted by $\mathbf{h}_{u,\text{fed}}^k$, which reflect the contribution of different devices to global model convergence.

Algorithm 1: Dual-Channel Spatio-Temporal Perception Encoding

Input: Device features $\{\mathbf{x}_m^k\}$, channel states $\{h_{m,u}^k, d_{m,u}^k\}$, divergence $\{(\delta_m^{k-1})^2\}$
Output: Embedding $\mathbf{E}_u^k$

for *each UAV* $u \in \mathcal{U}$ **do**
  Construct physical graph $\mathcal{G}_{\text{phy}}^k$ based on $\mathcal{N}(u)$;
  for *each device* $m \in \mathcal{N}(u)$ **do**
    Compute edge feature $\mathbf{e}_{m,u}^k \leftarrow [h_{m,u}^k, d_{m,u}^k]$;
  Aggregate spatial features: $\tilde{\mathbf{x}}_{u,\text{phy}}^k \leftarrow \text{ReLU}\left(\sum_{m\in\mathcal{N}(u)} \alpha_{m,u}^k \mathbf{W}_p[\mathbf{x}_m^k \parallel \mathbf{e}_{m,u}^k]\right)$;
  Temporal update via LSTM: $(\mathbf{h}_{u,\text{phy}}^k, \mathbf{C}_{u,\text{phy}}^k) \leftarrow \text{LSTM}(\tilde{\mathbf{x}}_{u,\text{phy}}^k, \mathbf{h}_{u,\text{phy}}^{k-1}, \mathbf{C}_{u,\text{phy}}^{k-1})$;
  Construct semantic graph $\mathcal{G}_{\text{fed}}^k$;
  Extract semantic feature: $\mathbf{h}_{u,\text{fed}}^k \leftarrow \text{GCN}(\mathcal{G}_{\text{fed}}^k)$;
  Compute fusion gate: $\mathbf{g}^k \leftarrow \sigma\left(\mathbf{W}_g[\mathbf{h}_{u,\text{phy}}^k \parallel \mathbf{h}_{u,\text{fed}}^k]\right)$;
  Fuse features: $\mathbf{E}_u^k \leftarrow \mathbf{g}^k \odot \mathbf{h}_{u,\text{phy}}^k + (1-\mathbf{g}^k) \odot \mathbf{h}_{u,\text{fed}}^k$;
return $\mathbf{E}_u^k$;

Finally, a gating mechanism is introduced to adaptively fuse the physical and semantic representations, yielding the unified spatio-temporal embedding

$$\mathbf{E}_u^k = \mathbf{g}^k \odot \mathbf{h}_{u,\text{phy}}^k + (1-\mathbf{g}^k) \odot \mathbf{h}_{u,\text{fed}}^k, \tag{20}$$

where $\mathbf{g}^k = \sigma(\mathbf{W}_g[\mathbf{h}_{u,\text{phy}}^k \parallel \mathbf{h}_{u,\text{fed}}^k])$ is a learnable gating vector that dynamically adjusts the relative contribution of each channel based on the current network state. The fused representation $\mathbf{E}_u^k$ is then used as the input to the policy

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network for subsequent decision-making, as summarized in Algorithm 1.

B. State Space and Action Space

In the proposed framework, each RL agent corresponds to one UAV $u \in \mathcal{U}$. The local observation state of UAV u at the k-th communication round is defined as $\mathbf{s}_u^k$, capturing both communication conditions and federated learning dynamics through a set of heterogeneous features, including the historical channel state sequence $\mathbf{H}_u^{k-\tau:k}$, the geometric distance vector $\mathbf{D}_u^k$ between devices and the UAV, the gradient divergence vector $\boldsymbol{\delta}_u^{k-1}$ reflecting data heterogeneity, and the remaining spectrum resource $\mathbf{B}_{\text{rem}}^k$. Accordingly, the state is expressed as

$$\mathbf{s}_u^k = \left(\mathbf{H}_u^{k-\tau:k}, \mathbf{D}_u^k, \boldsymbol{\delta}_u^{k-1}, \mathbf{B}_{\text{rem}}^k\right). \tag{21}$$

The action space comprises both discrete scheduling decisions and continuous resource-allocation variables, forming a hybrid space. Specifically, the discrete variable $s_{m,u}^k$ determines device association, while $b_{m,u}^k$ and f_m^k represent bandwidth allocation and CPU frequency control, respectively. To enable end-to-end learning in such hybrid spaces, a Gumbel–Softmax relaxation is adopted to approximate the discrete scheduling decisions. As $\tau_g \to 0$, the relaxed variable converges in distribution to a hard categorical sample, recovering the binary scheduling decision. In our implementation, τ_g is annealed from 1.0 to 0.05 over training, progressively sharpening the relaxation toward discrete decisions. The relaxed scheduling variable $\tilde{s}_{m,u}^k$ is formulated as

$$\tilde{s}_{m,u}^k = \frac{\exp\left((\rho_{m,u}^k + g_{m,u})/\tau_g\right)}{\sum_{j\in\mathcal{U}} \exp\left((\rho_{m,j}^k + g_{m,j})/\tau_g\right)}, \tag{22}$$

where $\rho_{m,u}^k$ denotes the scheduling logits generated by the Actor network, $g_{m,u}$ represents Gumbel noise, and τ_g is the temperature parameter.

For continuous resource allocation, Beta distributions are employed to model bounded decision variables. Specifically, the normalized bandwidth allocation coefficient $x_{m,u}^k$ and CPU allocation coefficient y_m^k are sampled as $x_{m,u}^k \sim \text{Beta}(\alpha_b, \beta_b)$, $y_m^k \sim \text{Beta}(\alpha_f, \beta_f)$, where $\alpha_b, \beta_b, \alpha_f, \beta_f$ are distribution parameters generated by the Actor network.

To ensure consistency between scheduling and resource allocation, a forward masking mechanism couples the two decisions: resource allocation is activated only when the corresponding device is selected, and suppressed to zero otherwise. At inference time, the binary scheduling decision is recovered via an argmax operation over the scheduling logits, which exactly satisfies constraints (C1) and (C2). This mechanism guarantees feasibility while preserving differentiability for policy learning.

C. Reward Function

Under the constrained Markov game formulation, each agent corresponds to one UAV $u \in \mathcal{U}$. The immediate reward of agent u at the k-th round is aligned with the system utility objective

$$r_u^k = \omega_1 \Upsilon_{FL}^k - \omega_2 E_{\text{total}}^k - \omega_3 T^k, \tag{23}$$

where $\omega_1, \omega_2, \omega_3 \geq 0$ are weighting coefficients that balance federated learning performance, energy consumption, and latency, respectively.

To account for the latency constraint (C6) and reliability constraint (C7), two instantaneous cost signals are introduced to quantify constraint violations

$$\begin{aligned} c_{u,1}^k &= \sum_{m\in\mathcal{M}} \max\left(0,\, s_{m,u}^k \left(T^k - T_{\max}\right)\right), \\ c_{u,2}^k &= \sum_{m\in\mathcal{M}} \max\left(0,\, s_{m,u}^k \left(\Gamma_{\min} - \Gamma_{m,u}^k\right)\right). \end{aligned} \tag{24}$$

Based on the above definitions, the policy optimization problem is formulated as a constrained reinforcement learning problem, where the objective is to maximize the expected cumulative reward while satisfying long-term average constraints

$$\begin{aligned} \max_{\pi} \quad & \mathbb{E}_\pi\left[\sum_{k=1}^{K} \gamma^k r_u^k\right] \\ \text{s.t.} \quad & \mathbb{E}_\pi\left[\sum_{k=1}^{K} \gamma^k c_{u,i}^k\right] \leq d_i, \quad i \in \{1,2\}, \end{aligned} \tag{25}$$

where π denotes the policy, $\gamma \in (0,1)$ is the discount factor, and d_i represents the tolerance threshold for each constraint. This formulation decouples system utility optimization from constraint satisfaction, progressively driving the system toward meeting the global accuracy constraint (C8).

D. Policy Optimization

To achieve efficient policy optimization under latency and reliability constraints, the FSTPO framework adopts a primal–dual proximal policy optimization (PPO) approach. The Actor network parameter θ_u is optimized by maximizing a clipped surrogate objective with constraint-aware advantage estimation

$$L_A(\theta_u) = \mathbb{E}_{\pi_{\theta_u}}\left[\min\left(\mu_u^k \hat{A}_{\text{total}}^k,\ \text{clip}(\mu_u^k, 1-\epsilon, 1+\epsilon)\hat{A}_{\text{total}}^k\right)\right], \tag{26}$$

where $\mu_u^k = \pi_{\theta_u}(\mathbf{a}_u^k \mid \mathbf{s}_u^k)/\pi_{\theta_{u,\text{old}}}(\mathbf{a}_u^k \mid \mathbf{s}_u^k)$ denotes the importance sampling ratio, and ϵ is the clipping parameter. The advantage terms are estimated using generalized advantage estimation (GAE).

To jointly account for system utility and constraint satisfaction, a weighted advantage function is introduced as

$$\hat{A}_{\text{total}}^k = \hat{A}_r^k - \lambda_1 \hat{A}_{c,1}^k - \lambda_2 \hat{A}_{c,2}^k, \tag{27}$$

where $\hat{A}_r^k$ denotes the reward advantage, $\hat{A}_{c,i}^k$ represents the cost advantage associated with the i-th constraint, and λ_i is the corresponding Lagrange multiplier.

The Lagrange multipliers are updated via projected gradient ascent with step size η_λ

$$\lambda_i^{(t+1)} = \max\left\{0,\ \lambda_i^{(t)} + \eta_\lambda\left(\mathbb{E}_\pi[c_{u,i}^k] - d_i\right)\right\}. \tag{28}$$

Under Slater's condition, this primal–dual iteration converges to a saddle point of the Lagrangian, guaranteeing asymptotic constraint satisfaction.

To improve convergence efficiency under dynamic network topologies, a topology-aware federated policy sharing mechanism is incorporated. Every E_{share} training episodes, the

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Algorithm 2: Federated Spatio-Temporal Policy Optimization (FSTPO)

Input: Episodes $E_{\max}$, horizon K, step sizes $\eta_\theta, \eta_\lambda$, clipping ϵ, sharing interval E_{share}
Output: Optimized Actor parameters $\{\theta_u\}$

Initialize Actor π_{θ_u}, Critic V_{ψ_u}, multipliers $\lambda_1, \lambda_2 \leftarrow 0$;
for $e = 1$ **to** $E_{\max}$ **do**
  for *each UAV* $u \in \mathcal{U}$ **do**
    Initialize state $\mathbf{s}_u^0$;
    for $k = 0$ **to** $K-1$ **do**
      Encode state $\mathbf{E}_u^k \leftarrow \text{Encoder}(\mathbf{s}_u^k)$;
      Sample action $\mathbf{a}_u^k \sim \pi_{\theta_u}(\cdot|\mathbf{E}_u^k)$;
      Execute action and observe r_u^k, $c_{u,1}^k$, $c_{u,2}^k$, $\mathbf{s}_u^{k+1}$;
      Store transition in $\mathcal{D}_u$;
    Compute advantages $\hat{A}_r^k$, $\hat{A}_{c,1}^k$, $\hat{A}_{c,2}^k$ via GAE;
    $\hat{A}_{\text{total}}^k \leftarrow \hat{A}_r^k - \lambda_1 \hat{A}_{c,1}^k - \lambda_2 \hat{A}_{c,2}^k$;
    Update Actor: $\theta_u \leftarrow \theta_u + \eta_\theta \nabla_{\theta_u} L_A(\theta_u)$;
    Update Critic: $\psi_u \leftarrow \psi_u - \eta_\theta \nabla_{\psi_u} L_V(\psi_u)$;
    Update multipliers: $\lambda_i \leftarrow \max\{0, \lambda_i + \eta_\lambda(\mathbb{E}[c_{u,i}^k] - d_i)\}$, $i \in \{1,2\}$;
    Clear buffer $\mathcal{D}_u$;
  if $e \bmod E_{\text{share}} == 0$ **then**
    Aggregate global policy: $\theta^{\text{glob}} \leftarrow \sum_{u\in\mathcal{U}} w_u \theta_u$;
    for *each UAV* $u \in \mathcal{U}$ **do**
      Soft update: $\theta_u \leftarrow \tau\theta^{\text{glob}} + (1-\tau)\theta_u$;
return $\{\theta_u\}$;

Actor parameters across UAVs are aggregated via weighted model averaging, where aggregation weights are determined by current network connectivity. The aggregation weight is $w_u = \sum_m s_{m,u}^k / \sum_{i\in\mathcal{U}} \sum_m s_{m,i}^k$, proportional to each UAV's active device count. Each UAV then performs a soft update to integrate the global policy with its local policy, balancing global consistency and local adaptability. The overall procedure is summarized in Algorithm 2.

E. Complexity Analysis

The computational complexity of Algorithm 1 per UAV per round is $\mathcal{O}(|\mathcal{N}(u)| \cdot d_h \cdot (d_x + d_e) + |\mathcal{N}(u)|^2 \cdot d_h + d_h^2)$, where d_h is the hidden dimension and d_x, d_e denote the node and edge feature dimensions, respectively. Since all UAV agents execute Algorithm 1 in parallel, the total per-round encoding complexity scales as $\mathcal{O}(M^2 d_h/U)$ under uniform device distribution, confirming that the decentralized architecture reduces per-agent complexity linearly with the number of UAVs. For Algorithm 2, the Actor update per agent per episode requires $\mathcal{O}(N_{\text{epoch}} \cdot B \cdot d_h^2)$ operations, which is approximately two orders of magnitude smaller than the FL simulation cost per episode and contributes less than 10% of total training time. The overall per-episode training complexity is therefore dominated by the FL simulation, scaling as $\mathcal{O}(K \cdot M^2 d_h + U \cdot N_{\text{epoch}} \cdot B \cdot d_h^2)$, which grows linearly with the number of communication rounds K and UAVs U.

Table I: Simulation Parameters

Parameter	Value
Service area	1000×1000 m^2
Number of UAVs U	$3 \sim 6$
UAV altitude H_u	$80 \sim 120$ m
Number of devices M	$10 \sim 60$
Bandwidth capacity $B_{\max}$	$15 \sim 25$ MHz
Transmit power P_m	$20 \sim 24$ dBm
Noise spectral density N_0	-174 dBm/Hz
Channel gain β_0	-30 dB
Model size Z	$2.0 \sim 5.5$ MB
Task size v_m^k	$1.0 \sim 5.0$ MB
Device CPU frequency f_m	$0.5 \sim 2.5$ GHz
UAV aggregate throughput f_u	$8 \sim 12$ GHz
CPU cycles c_m, c_u	$800 \sim 1200$ cycles/bit
Capacitance coefficient κ_m, κ_u	10^{-28}
Max delay $T_{\max}$	5 s
SNR threshold $\Gamma_{\min}$	5 dB

IV. Experimental Settings

To evaluate the performance of the proposed FSTPO framework, a three-dimensional air–ground collaborative MEC scenario is constructed. The target service area is set to 1000×1000 m^2. The number of UAVs U is randomly selected from $[3, 6]$, and their flight altitudes H_u are uniformly distributed within $[80, 120]$ m. The number of ground devices M varies within $[10, 60]$. The initial positions of devices are uniformly distributed over the target area and evolve according to a random walk mobility model during communication. To enhance the generality of the simulation results, all communication and computation parameters are independently sampled from predefined intervals unless otherwise specified. The detailed parameter settings are summarized in Table I [7], [12], [19], [33]. Each experiment is repeated multiple times, and the results are averaged.

The experiments are conducted on the Fashion-MNIST and CIFAR-10 datasets for image classification. The input dimensions are 28×28 grayscale images and 32×32 colour images, respectively. A lightweight convolutional neural network (CNN) is adopted as the global model, trained using the cross-entropy loss function. To model data heterogeneity, the dataset is partitioned across devices according to a Dirichlet distribution with parameter $\alpha \in [0.3, 0.7]$, yielding non-IID local datasets. This setting induces heterogeneous data distributions across devices, which is consistent with the gradient divergence metric used in the proposed framework. The total number of global communication rounds is set to $K = 100$, and the number of local training iterations is randomly selected from $[3, 7]$. The batch size is 64, and the learning rate is $\eta = 0.01$. The target accuracy thresholds are set to 85% for Fashion-MNIST and 75% for CIFAR-10, calibrated to represent approximately 90% of the centralized training accuracy ceiling under the adopted model architecture and dataset partitioning scheme, ensuring the accuracy constraint

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is neither trivially satisfiable nor infeasible within $K = 100$ rounds.

Under the constrained Markov game framework, each UAV independently runs an Actor–Critic architecture trained using the Adam optimizer. The learning rates of the Actor and Critic networks are set to 10^{-4} and 3×10^{-4}, respectively. The temporal window length is $\tau = 3$, and the LSTM hidden dimension is 128. The Gumbel–Softmax temperature is annealed from 1.0 to 0.05 over training. The PPO clipping parameter is $\epsilon = 0.2$, and the discount factor is $\gamma = 0.99$. The Lagrange multiplier step size is $\eta_\lambda = 0.05$. The utility weights are set as $\omega_1 = 1$, $\omega_2 = 0.1$, and $\omega_3 = 0.5$, calibrated to bring the three utility terms into a comparable numerical scale given their differing physical units. The federated policy sharing interval is $E_{\text{share}} = 5$, the soft update coefficient is $\eta_f = 0.01$, and the maximum number of training episodes is $E_{\max} = 2000$.

For fair comparison, all baseline methods are implemented under the same system settings, including identical state information, action constraints, and resource limitations. The following baseline algorithms are considered for comparison:

1) *MADDPG*: A multi-agent deep deterministic policy gradient method that outputs continuous resource allocation variables $b^k_{m,u}$ and f^k_m, with heuristic thresholding applied for discrete scheduling decisions.
2) *Greedy*: Devices are associated with UAVs based on instantaneous channel gains, and bandwidth resources are equally allocated among connected devices.
3) *PPO*: A standard proximal policy optimization method without constraint handling, where a multilayer perceptron replaces the proposed graph-based encoder, and no federated policy sharing is employed.
4) *Random*: Scheduling and resource allocation decisions are randomly generated within the feasible region defined by system constraints.
5) *FedAvg*: A classical federated learning approach where a fixed number of devices are randomly selected in each round, with equal bandwidth allocation.

V. Experimental Results

To comprehensively evaluate the performance of the proposed FSTPO framework, we categorize the experimental analysis into five dimensions: convergence dynamics, learning robustness under statistical heterogeneity, resource efficiency and scalability, utility weight sensitivity, and a detailed ablation study of the core algorithmic components.

A. Convergence and Optimization Dynamics

The primary objective of the proposed framework is to maximize system utility while strictly adhering to delay constraints. The expected utility convergence and the instantaneous delay violation costs, alongside the Lagrange multiplier dynamics, are presented in Fig. 3. FSTPO consistently achieves the highest utility plateau, demonstrating relative utility improvements of 24.26%, 16.37%, 63.14%, and 125.21% over MADDPG, PPO, Greedy, and FedAvg, respectively. This gain stems from the Gumbel–Softmax and Beta distribution

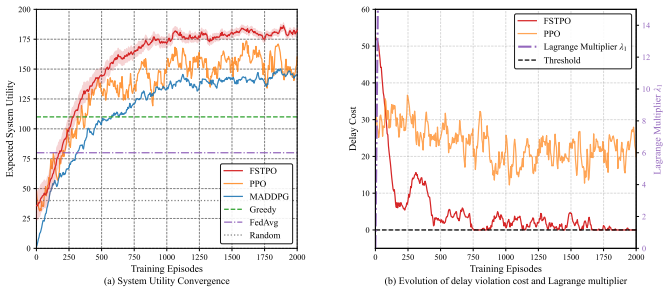


Fig. 3: Convergence of system utility and delay violation cost.

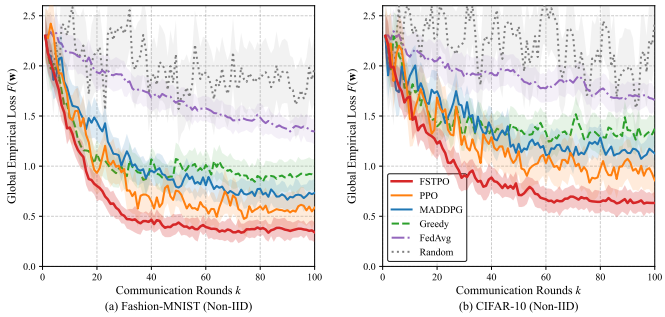


Fig. 4: Global empirical loss under different policies.

decoupling, which avoids the policy oscillations induced by rigid discretization in MADDPG. The constraint-handling efficacy is confirmed in Fig. 3(b): while PPO sustains a continuous delay violation above 5, FSTPO's primal–dual mechanism drives λ_1 to penalize violations and restore feasibility, with the violation cost converging to 0.

This stable optimization trajectory is further corroborated by the global empirical loss $F(\mathbf{w})$ depicted in Fig. 4. FSTPO reaches the minimum loss with the tightest standard deviation. A notable crossover effect is observed in the Greedy policy: despite exhibiting the steepest initial loss reduction by aggressively exploiting devices with excellent physical channel conditions, its trajectory severely stagnates and rebounds. This confirms the destructive impact of gradient divergence $(\delta^{k-1}_m)^2$; persistently aggregating highly Non-IID data ultimately causes the global model to deviate from the true optimum. In contrast, FSTPO ensures a smooth descent by semantically isolating divergent data, successfully mitigating the high-frequency jitter prevalent in standard DRL baselines.

B. Learning Performance and Robustness to Heterogeneity

Test accuracy on held-out data is a key metric for evaluating federated learning systems in heterogeneous environments. The evolution of test accuracy on the Fashion-MNIST and CIFAR-10 datasets is shown in Fig. 5. FSTPO is the sole algorithm that consistently satisfies the target accuracy limits, namely $Acc_{\text{target}} = 85.0\%$ and 75.0% respectively, within 100 communication rounds. On the CIFAR-10 dataset, the performance gain becomes particularly pronounced, yielding relative accuracy improvements of up to 40.36% over the baselines. By encoding data heterogeneity via the semantic graph channel into $\mathbf{E}^k_u$, FSTPO filters devices with severe

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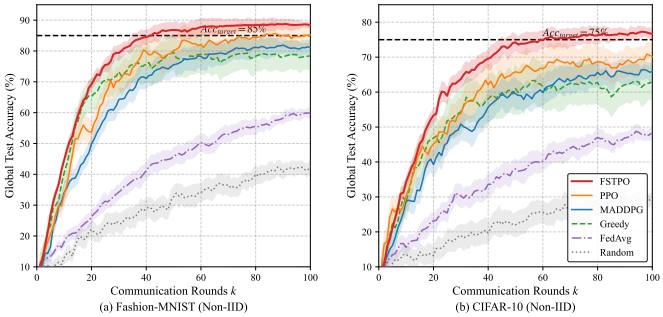


Fig. 5: Global test accuracy on Non-IID datasets.

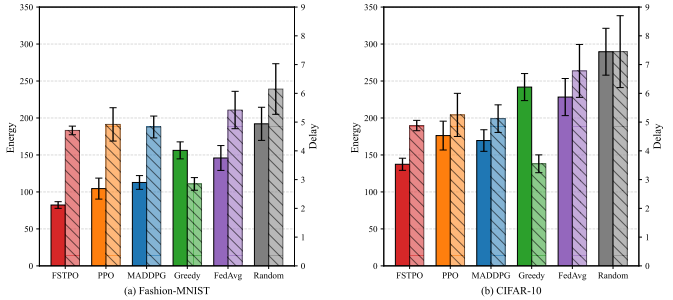


Fig. 7: Average per-round energy consumption and task latency per communication round.

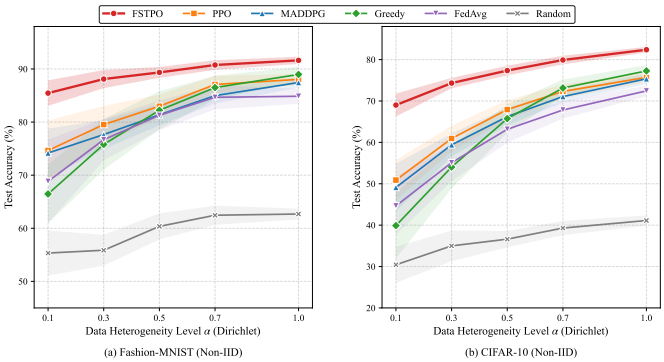


Fig. 6: Test accuracy under varying data heterogeneity α.

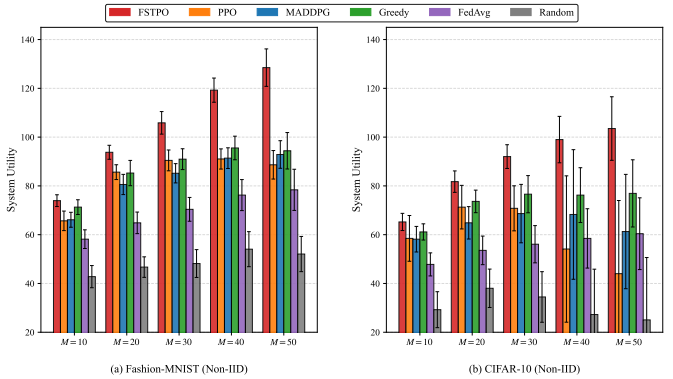


Fig. 8: System utility versus the number of scheduled devices.

distribution skews, yielding superior accuracy and consistently narrow variance across rounds.

To stress test the system's structural capacity against data skewness, performance under varying degrees of statistical heterogeneity, parameterized by the Dirichlet concentration parameter α, is evaluated in Fig. 6. In extreme Non-IID regimes, such as when $\alpha = 0.1$, the channel-heuristic Greedy policy suffers a catastrophic accuracy collapse due to its myopic prioritization of physical channel states over local data quality. Standard DRL and FedAvg similarly fail to maintain convergence. In contrast, FSTPO's graph-structured encoding captures implicit divergence signals, balancing resource constraints with data coverage and maintaining accuracy across all heterogeneity levels.

C. Resource Efficiency and System Scalability

Practical deployment requires a delicate balance among communication reliability, energy consumption, and system scalability. The average energy consumption and task latency across different workloads are compared in Fig. 7. FSTPO establishes a superior Pareto-like balance, maintaining the lowest energy consumption among all compared schemes, with per-round reductions of approximately 19.3% and 34.7% relative to PPO and Greedy, respectively, in the CIFAR-10 scenario. While the Greedy approach minimizes latency, it incurs an excessively high energy penalty, illustrating a disproportionate trade-off. As task complexity increases, the incremental overhead of FSTPO remains marginal compared to baselines like FedAvg and Random, which demonstrate insufficient adaptability to dynamic environments.

The framework's scalability is validated in Fig. 8, which plots system utility as the active device pool expands from 10 to 50. FSTPO not only maintains the highest utility but also exhibits proportional growth, indicating highly efficient utilization of supplemental computing and communication resources. In contrast, standard PPO and MADDPG experience severe growth stagnation and policy collapse under high-load conditions, particularly on the CIFAR-10 dataset, evidenced by significantly widened error bars.

Finally, the system resilience under physical channel impairments is evaluated by varying the decoding SNR threshold $\Gamma_{\min}$ in Fig. 9. As $\Gamma_{\min}$ increases, communication conditions become more demanding, and the success probability of all schemes declines accordingly. The degradation is especially evident for DRL-based baselines such as PPO and MADDPG, whose performance drops rapidly in the high-threshold regime, suggesting that without explicit reliability constraint awareness, the learned policies allocate resources unsuitably for harsh channel conditions. The Greedy strategy maintains a high success rate by significantly reducing participating devices, indicating a conservative scheduling pattern that limits system coverage. In contrast, FSTPO exhibits a more balanced response: the success probability remains above the reliability requirement even under severe fading, while a relatively larger set of active devices is preserved throughout, reflecting more efficient use of system resources.

To further examine robustness beyond average performance, a statistical and sensitivity analysis is conducted under dynamic network conditions, as illustrated in Fig. 10. The

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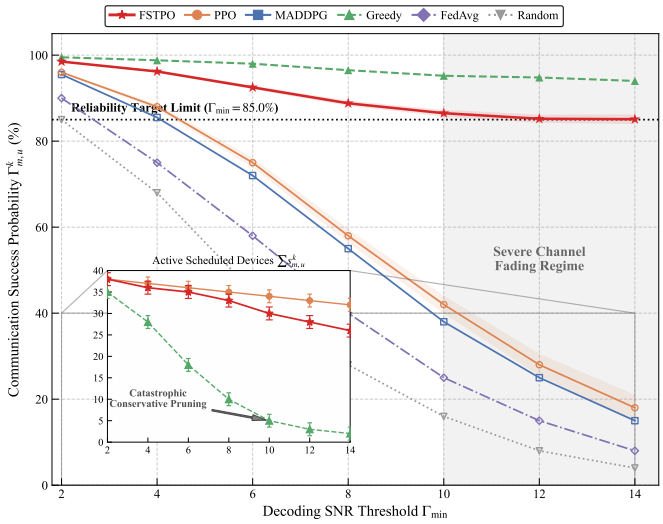


Fig. 9: System performance versus SNR decoding thresholds $\Gamma_{\min}$.

cumulative distribution functions of task latency and energy consumption in Fig. 10(a) and (b) show that FSTPO yields a tightly concentrated latency distribution, where all tasks are completed within $T_{\max} = 5$ s. In comparison, PPO and MADDPG exhibit evident long-tail behaviours, indicating frequent delay violations. The Greedy strategy achieves relatively low latency at the expense of substantially higher energy consumption. FSTPO maintains low latency dispersion while preserving energy efficiency, suggesting a more balanced operating regime. The impact of the gradient penalty coefficient λ on learning stability is presented in Fig. 10(c). A small λ leads to noticeable oscillations during training, whereas an excessively large value restricts device participation and slows convergence. A moderate setting $\lambda = 1.0$ yields the most stable training trajectory. The effect of the local iteration number I is illustrated in Fig. 10(d). As I increases, computational cost rises steadily while system utility first improves then gradually declines, indicating the presence of a balanced operating region where moderate local computation achieves an effective trade-off between learning performance and resource consumption.

D. Utility Weight Sensitivity Analysis

To evaluate the robustness of the proposed framework to the choice of utility weighting coefficients, a sensitivity analysis is conducted by independently varying ω_2 and ω_3 around the baseline setting $(\omega_1, \omega_2, \omega_3) = (1, 0.1, 0.5)$, as illustrated in Fig. 11. When ω_2 is reduced to 0.05, the policy assigns lower priority to energy conservation, resulting in higher energy consumption but improved learning accuracy. Conversely, increasing ω_2 to 0.2 drives the policy toward more aggressive energy reduction at a moderate accuracy cost, yet both configurations continue to satisfy the target accuracy thresholds of 85.0% and 75.0% on Fashion-MNIST and CIFAR-10, respectively. When ω_3 is varied between 0.1 and 1.0, the energy consumption exhibits a monotonically increasing trend with ω_3, as larger latency weights incentivize higher CPU frequency allocations that reduce round duration at the cost of greater power consumption, while accuracy remains largely stable across both datasets. Importantly, the latency constraint $T^k \leq T_{\max} = 5$ s is satisfied across all weight configurations, confirming that the Lagrange multiplier mechanism enforces hard constraints independently of the utility coefficient selection. These results demonstrate that FSTPO is robust to moderate variations in (ω_2, ω_3) and that the baseline coefficient setting represents a well-balanced operating point within the feasible Pareto region.

E. Ablation Study

To evaluate the contribution of each component in the proposed architecture, an ablation study is conducted with three degraded variants: w/o SemGraph, which removes semantic-aware modelling; w/o TFPS, which disables policy sharing across agents; and w/o ContRelax, which replaces continuous action relaxation with direct discretization. As shown in Fig. 12, the full FSTPO framework maintains a smooth and stable convergence trend on both Fashion-MNIST and CIFAR-10, reaching a higher accuracy level with relatively small fluctuations.

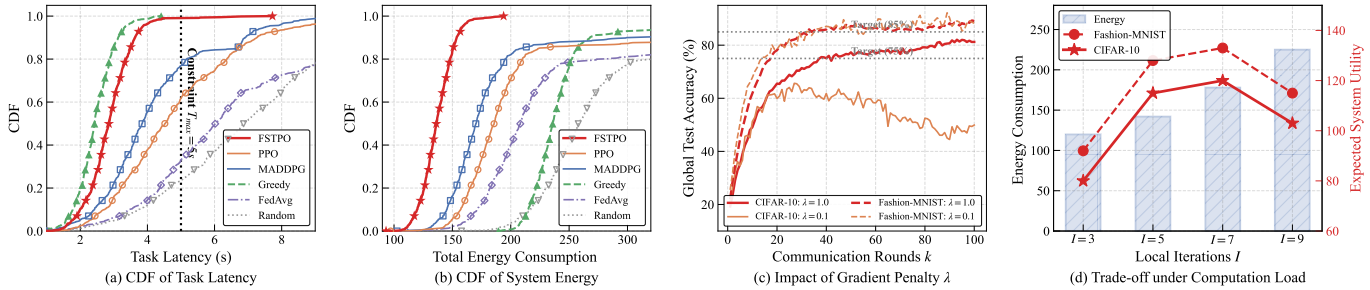
The w/o TFPS variant shows a slower rate of increase in accuracy and lower final performance. The gap becomes more apparent as training proceeds, indicating that removing policy sharing weakens cooperation among distributed agents and reduces the system's ability to adapt to temporal variations. The w/o SemGraph variant shows an increasing trend with multiple abrupt drops, especially on CIFAR-10, as heterogeneous updates are aggregated without divergence filtering, causing intermittent model degradation. The w/o ContRelax variant exhibits continuous oscillations during training. The accuracy curve fluctuates at a high frequency and fails to stabilize, remaining at a comparatively lower level. This reflects that direct discretization disrupts the smoothness of the action space and introduces instability into the optimization process, confirming the necessity of the Gumbel–Softmax relaxation for effective hybrid action learning.

VI. Conclusion

This paper proposed FSTPO, a federated spatio-temporal policy optimization framework for UAV-assisted edge computing. By modeling the resource management problem as a constrained Markov game and incorporating dual-channel spatio-temporal state encoding with hybrid action learning, FSTPO jointly optimizes device scheduling, bandwidth allocation, and CPU frequency control under Non-IID data distributions. Simulation results show that FSTPO achieves 24.26%–125.21% utility gains over baselines, satisfies latency and reliability constraints via primal–dual optimization, and maintains stable convergence across varying heterogeneity levels. Future work will extend the framework to incorporate UAV trajectory design and more realistic mobility models.

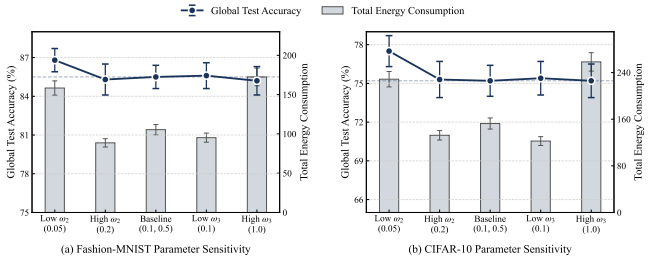
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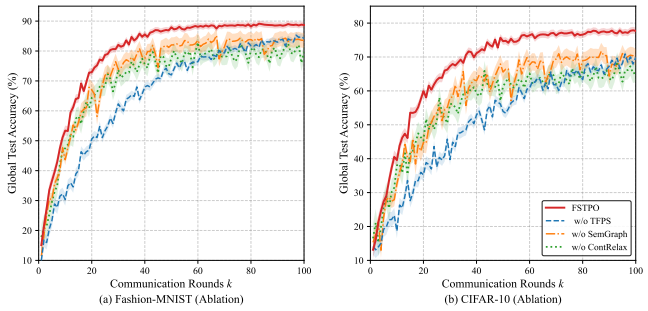
(a) CDF of Task Latency (b) CDF of System Energy (c) Impact of Gradient Penalty λ (d) Trade-off under Computation Load

Fig. 10: Statistical performance and sensitivity analysis.



(a) Fashion-MNIST Parameter Sensitivity (b) CIFAR-10 Parameter Sensitivity

Fig. 11: Sensitivity analysis of utility weighting coefficients ω_2 and ω_3.



(a) Fashion-MNIST (Ablation) (b) CIFAR-10 (Ablation)

Fig. 12: Ablation study of the proposed framework.

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